

Hanoi Urban Noise Mapping

Data Collection Report

363 smartphone measurements · 3 sites · 10 Jun to 22 Jul 2026

363

Measurements

3

Sites

47-88

L_{A,25s} range

0.14

Model R²

39%

> QCVN day

1. Objective

Map and characterise urban noise in three contrasting districts of Hanoi, and predict noise from urban morphology. We follow the Sunbird AI methodology (Urban Noise Uganda 61K, Nsumba et al., 2026): smartphone field collection plus a morphology-to-noise model, reproduced then applied here.

Measurement status: our target is a 20-30 s A-weighted level from consumer smartphones (denoted L_{A,25s}), not a certified L_{Aeq}. The three phones are cross-calibrated against each other, never against a reference structure. As only 100 hours of data are supported, ABSOLUTE levels are indicative (plausible bias +3.0 to +7.3 dB, section 3.4). No compliance claim is made anywhere in this report.

2. Data Collection summary

Site	n	Median dB	Min-Max	% < 60 dB	% roadside
Ocean Park	184	68	47-88	28%	73%
Hoan Kiem lake	99	70	55-80	7%	89%
Vinh Tuy area	80	66	53-76	20%	74%
ALL	363	68	47-88	21%	78%

Zones: Ocean Park = new high-rise development (shielding, construction); Vinh Tuy = transport infrastructure (heavy traffic); Hoan Kiem = historic old quarter (narrow streets, pedestrian zones). QC: site labels cross-checked against GPS (6 mislabelled submissions reassigned).

Headline findings: median 68 dB; 39% of short samples sit above the QCVN daytime threshold value; levels peak at 18:00. A model trained directly on our data reaches R² 0.14 [-0.10, 0.33] under buffered loo 300 m (section 5), against baselines reported in the same table.

3. Methodology

3.1 Field data-collection protocol

Study zones were mapped in Google Earth and chosen for contrasting typologies (section 2). At each point we recorded a 20-30 s reading (with a 10 s audio clip) in ODK Collect, submitted live to a KoboToolbox server with automatic timestamp and GPS. Phones were held at 1.2 m (SPB positioning) at varied distances from the road, by three cross-calibrated collectors, from 05:00 to 23:00 with emphasis on rush hours. Weather (Open-Meteo) and spatial context (OSM) are added.

3.2 Reproduction of the Sunbird pipeline

We reproduced the full Sunbird chain on their Uganda dataset (notebooks 01-06): cleaning, audio QC, morphology features, figures, surrogate model. Our statistics match the paper (median 49 dB vs their 45-50 dB).

3.3 Model features

Morphology within 300 m of each point (OpenStreetMap): built-area ratio, road density, intersection count, distance to nearest road; plus hour of day and weekend flag. Model: LightGBM.

3.4 Reference values and measurement status

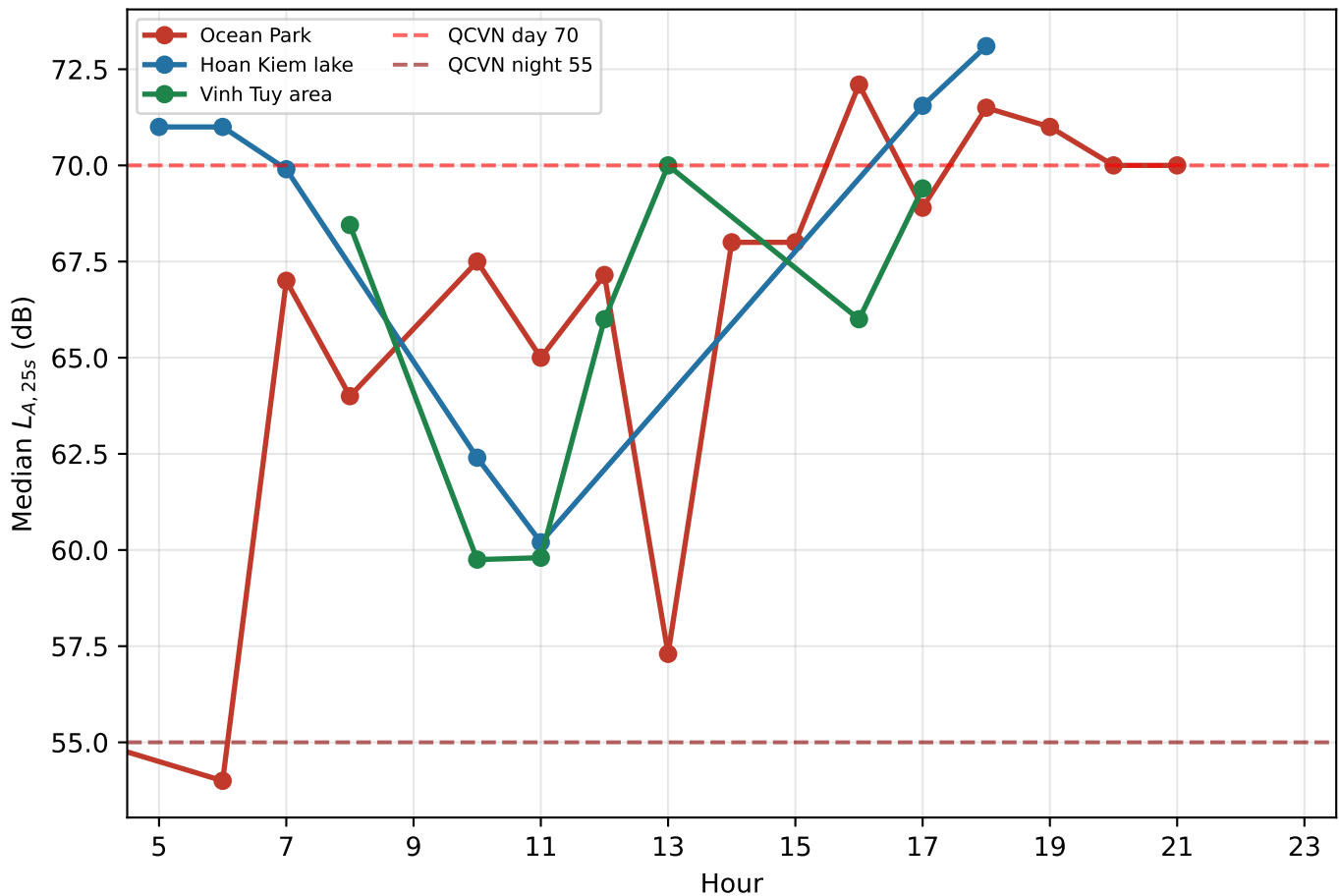
Reference value	Day (6-21h)	Night (21-6h)
QCVN 26:2010/BTNMT (Vietnam, ordinary area)	70 dB	55 dB

The WHO road-traffic guideline values (53 / 45 dB) have been WITHDRAWN from this report. They are L_{den} and L_{night}: ANNUAL averages with +5 dB evening and +10 dB night penalties. Comparing them to a 25 s daytime sample compares different statistics of different processes. QCVN is retained because it regulates a level, but it regulates an L_{Aeq} measured with a class 1-2 meter under TCVN 7878-2:2010: our instrument does not meet that specification either.

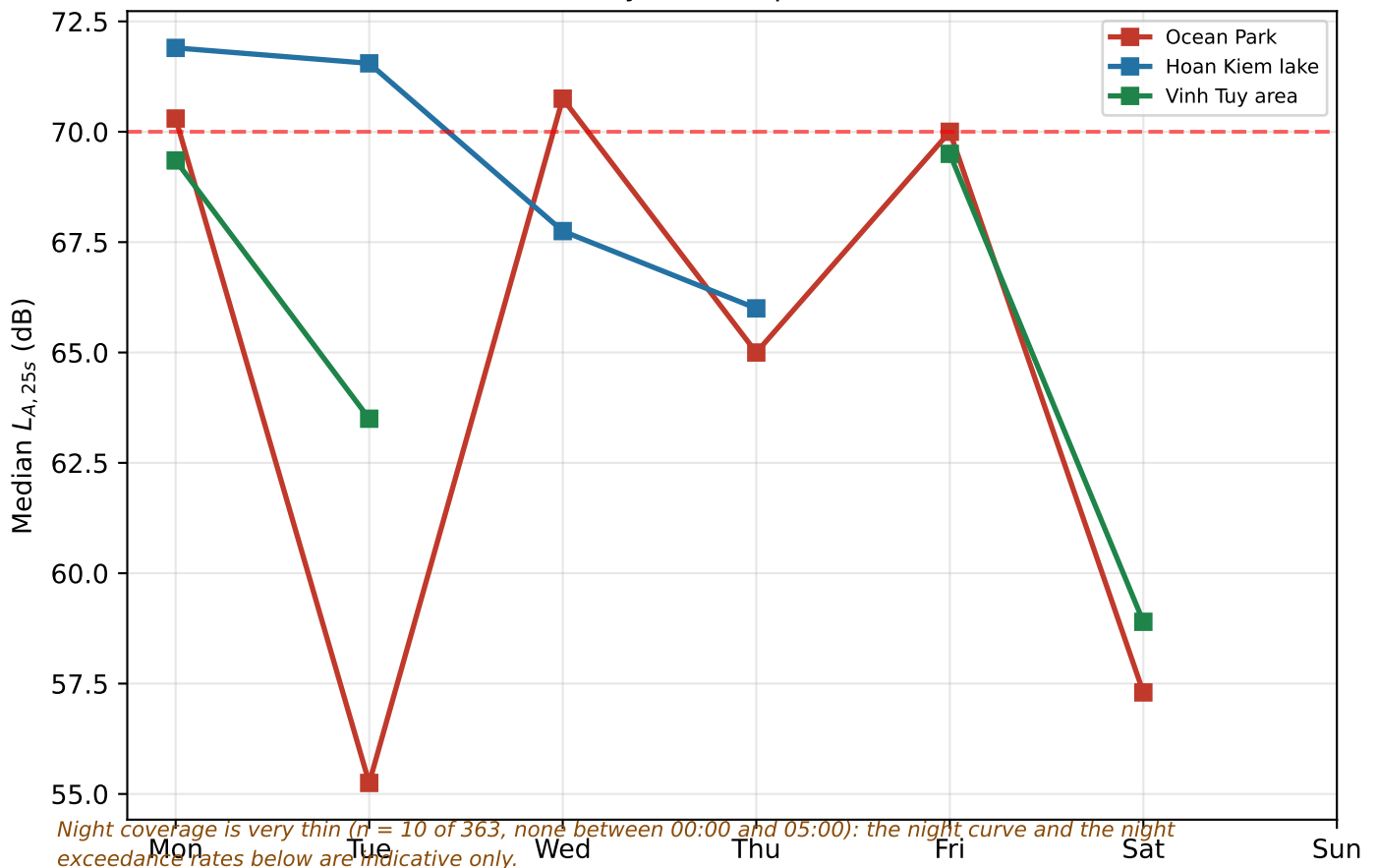
Consequence for every exceedance figure in this report: it is a DESCRIPTIVE STATISTIC of our sample - the share of short samples above a threshold value - and never a finding of regulatory non-compliance. Sensitivity to the calibration bias (+3.0 to +7.3 dB) is given on page 4.

4. Data analysis: temporal patterns

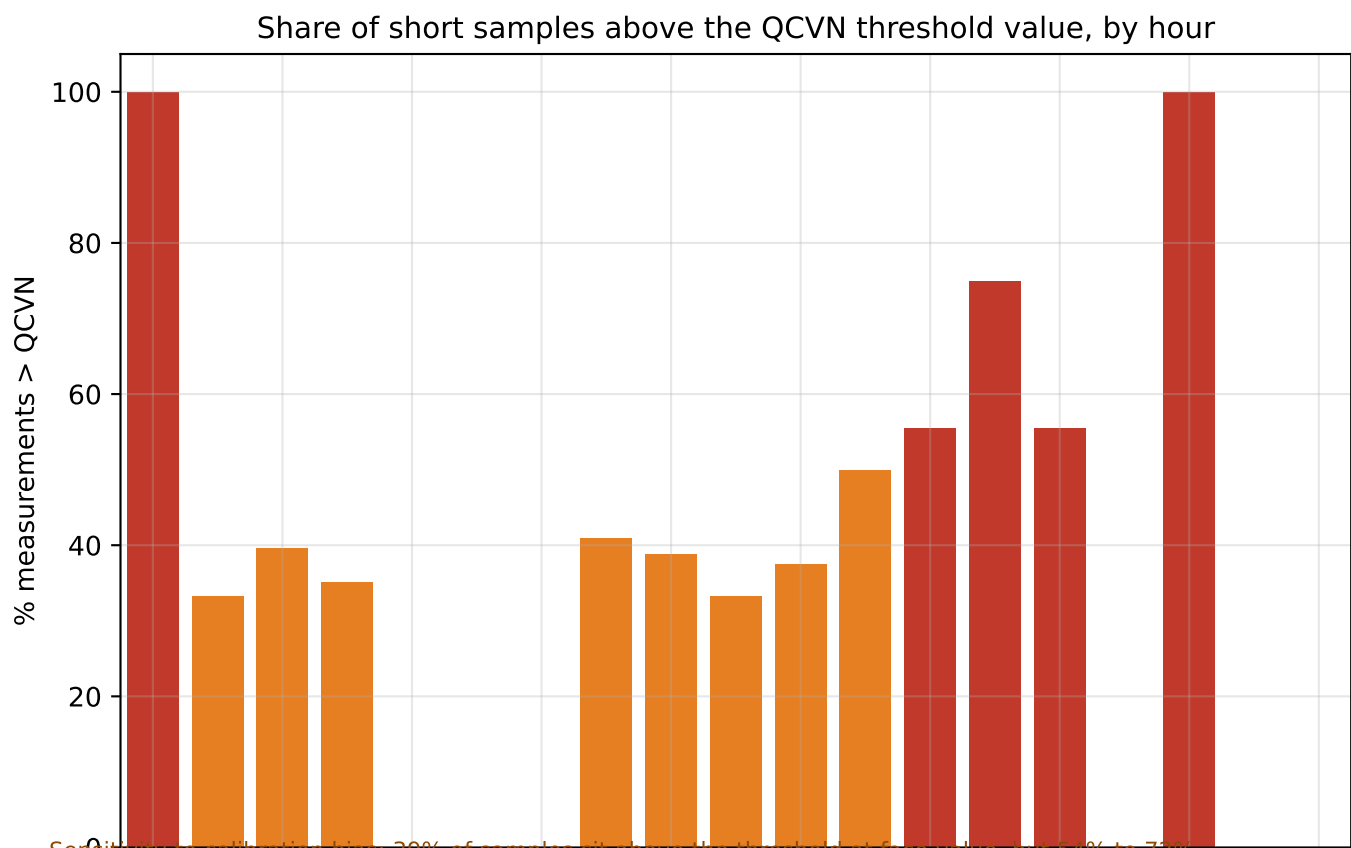
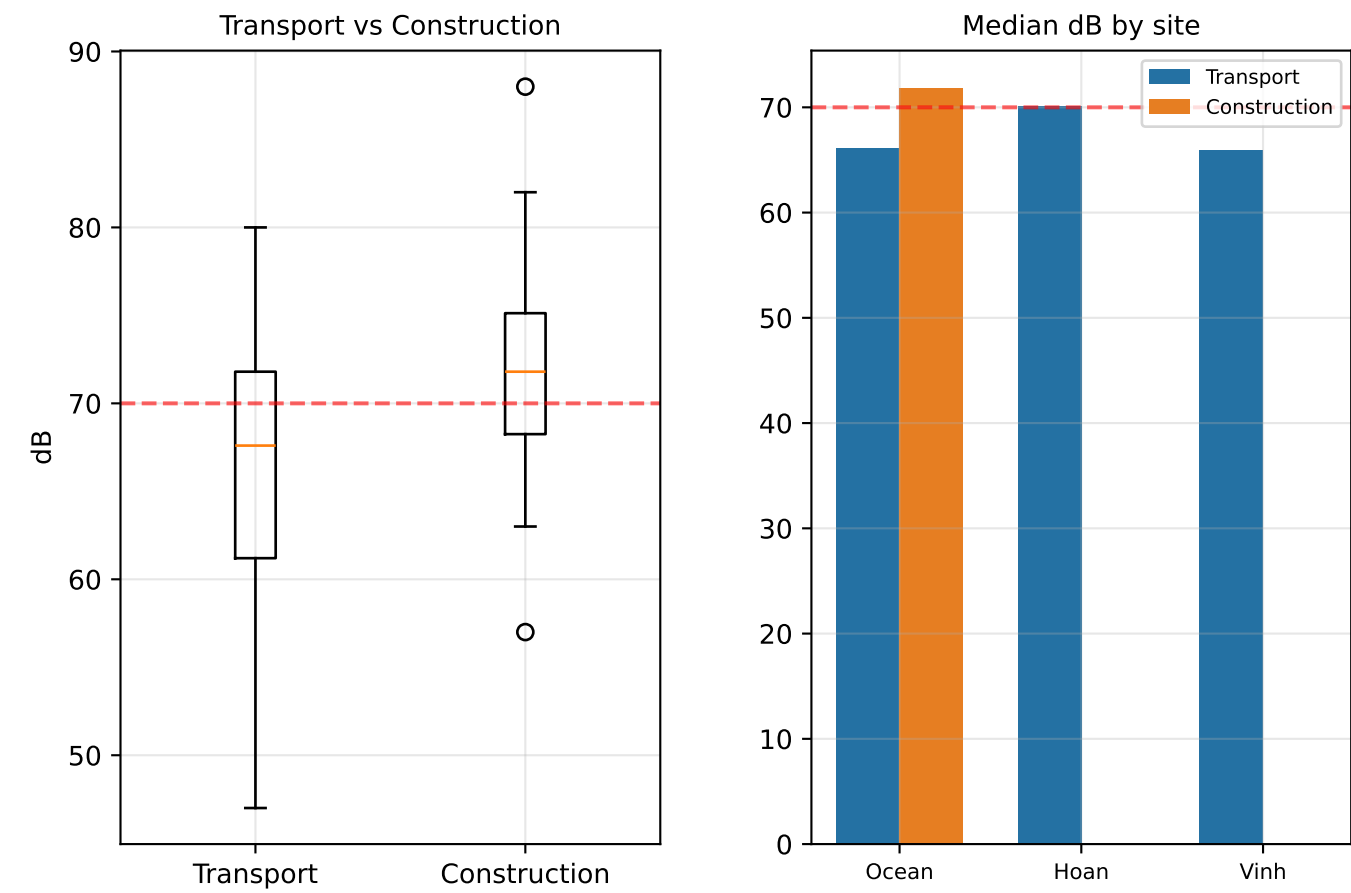
Hourly pattern within a day (capture window 05:00-23:00)



Day-of-week pattern



4. Data analysis: sources & exceedances

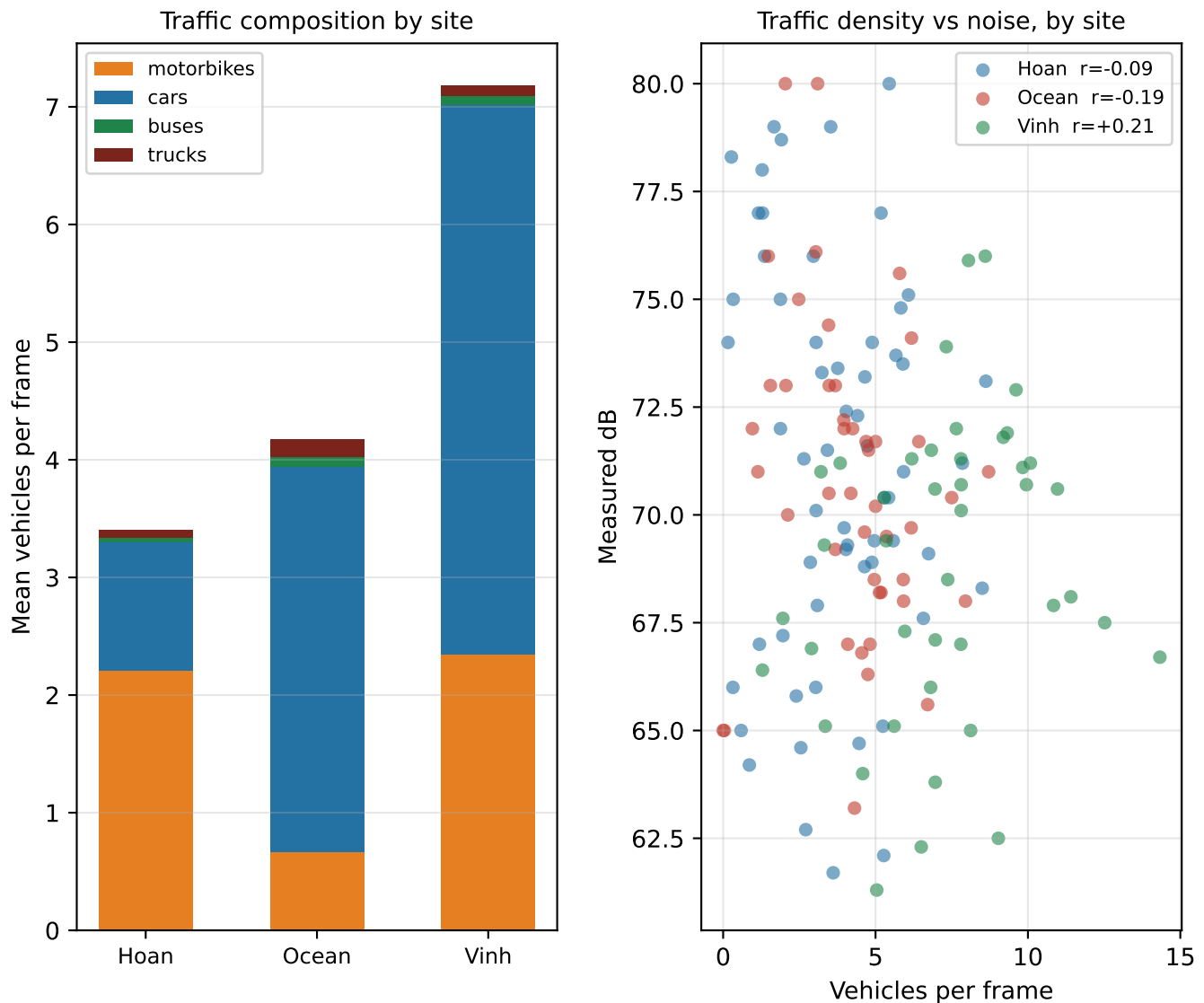


Sensitivity to calibration bias. 39% of samples sit above the threshold at face value, but 64% to 72% once the plausible instrumental bias (+3.0 to +7.3 dB, section 3.4) is applied. The exposure PATTERN - which hours and which sites are loudest - is unaffected by a common bias; the exceedance LEVEL is. We report the pattern as a finding and the level as an indication.

4. Data analysis: traffic videos & weather

147 traffic videos (~25 s each) matched to their measurement by timestamp (median gap 15 s).

Vehicles counted with YOLOv8 at ~1 frame/s (3726 frames analysed): average vehicles visible per frame.



Findings: traffic composition matches the zone typologies: Hoan is motorbike-dominated (65%), Ocean car-dominated (79%), Vinh the densest corridor (up to 14 vehicles/frame).

Vehicle density does NOT track noise: the site-wise correlations are weak and of inconsistent sign (Hoan -0.09, Ocean -0.19, Vinh +0.21), negative on 2 of 3 sites, and -0.15 overall. Density is a stock; emission is driven by flow and speed, which a per-frame count cannot observe - see section 7. Note: small motorbikes are harder to detect than cars and parked vehicles are counted too (no motion filter yet), so motorbike shares are a lower bound, and the detector was never validated against manual counts.

Weather: no robust effect. Raw correlations (temperature +0.26, rain -0.29) vanish once hour of day and session are controlled: quiet-point campaigns happened to fall on rainy days. Wind shows no effect on readings (no microphone artefact).

5. Predictive model

LightGBM mapping urban morphology (300 m radius, OpenStreetMap) plus time of day to level.

5.1 How performance is measured (corrected August 2026)

Earlier versions of this report grouped cross-validation on ~110 m cells. The model features are aggregates over a 300 m RADIUS, so two points 110 m apart share more than 85% of their disc: the model saw near-twins of its test points and the reported score was not out-of-sample. That protocol is replaced by buffered loo 300 m, and every model below - including the baselines - is scored on exactly the same splits, with bootstrap confidence intervals resampled by spatial block.

5.2 Model comparison - Buffered LOO 300 m

Model	R ²	95% CI	MAE (dB)	r
Moyenne globale	-0.04	[-0.12, -0.03]	6.06	-0.44
Moyenne par site	-0.20	[-0.55, -0.03]	6.62	-0.06
Moyenne par (site, heure)	-0.42	[-0.77, 0.03]	6.58	0.06
Régression sur log(distance route)	0.20	[0.08, 0.27]	5.28	0.45
Distance inverse (k=8, p=2)	-0.20	[-0.37, 0.02]	6.44	0.02
LightGBM — temps seul (ablation)	0.08	[-0.13, 0.46]	5.19	0.39
LightGBM — morphologie seule (ablation)	0.04	[-0.24, 0.23]	5.63	0.41
LightGBM — morphologie + temps (modèle du projet)	0.14	[-0.10, 0.33]	5.26	0.47

Key figure - a one-variable physical baseline wins. An OLS regression on log(distance to road) scores R² 0.200 against 0.137 for the six-variable LightGBM: morphology aggregated over 300 m adds nothing over the distance term alone (morphology-only ablation: 0.041). Against a (site, hour) lookup table the full model still gains dR² = +0.556 / dMAE = +1.32 dB - but that gain is carried by TIME, not space.

5.3 Generalisation to an unseen typology (leave-one-site-out)

Test site	n	R ²	MAE (dB)
Ocean Park	184	0.19	5.80
Hoan Kiem lake	99	-0.42	5.30
Vinh Tuy area	80	-0.58	5.85

Each site is predicted by a model trained on the other two only. A negative R² means: worse than predicting the global mean everywhere. With three sampled typologies, the number of independent morphological configurations is close to three, whatever the number of points: the model INTERPOLATES within sampled typologies, it does not EXTRAPOLATE. Maps are clipped to the sampled envelope.

Cross-city transfer (Uganda -> Hanoi) is reported in section 7 as a methodological result, not as a method. No comparison is made with published scores from other cities: they target a different quantity (long-term L_{Aeq}).

6. Agent-based simulation (GAMA)

The predicted noise map is loaded into an agent-based model with moving vehicles, construction sites and interactive scenario controls.

6.1 What the simulation contains

Layer	Source	Status
Background noise level, per hour	LightGBM on our 363 measurements	predicted
Traffic density and vehicle mix	147 videos, YOLO counts	measured
Construction excess (+2 dB at 56 m)	our 32 near-site measurements	calibrated
Vehicles as sound sources	not identifiable in our data	excluded
Traffic-volume scenario	10 log10 of the volume factor	physical law

6.2 Scenario controls

Hour of day (5h-21h, each hour predicted separately, vehicle numbers follow the traffic measured at that hour) · traffic volume (x0.2 to x3) · construction on/off with working hours · mitigation: 30 km/h zone (-3 dB) or pedestrianisation (traffic x0.2).

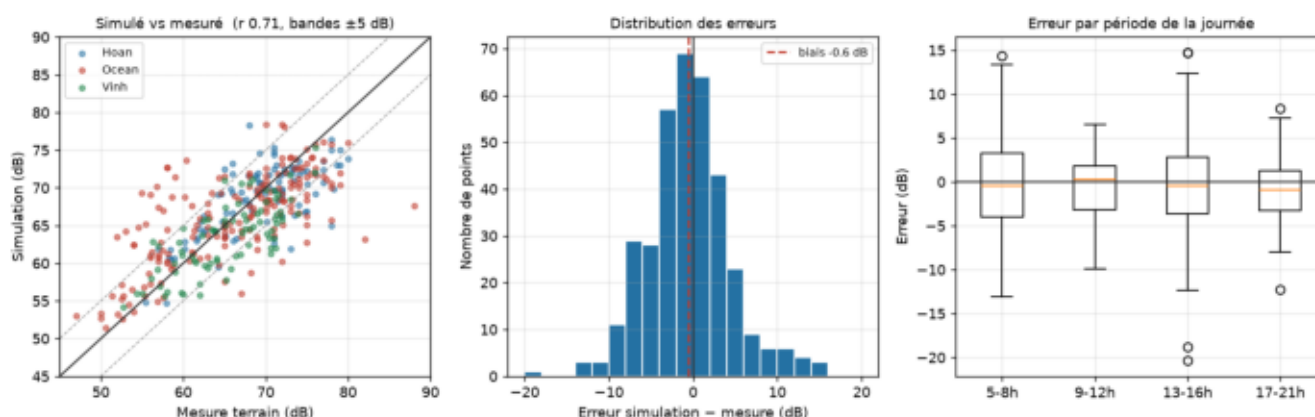
Calibration note: we tried to derive a per-vehicle emission level from our own data (non-negative energy regression on the 147 matched videos). All three coefficients came out at zero - at a given site, the number of visible vehicles does not explain the measured level (R2 0.008-0.042). Rather than inject invented values, vehicles are displayed as a calibrated picture of the fleet but carry no sound in the computation.

6.3 Does the simulation reproduce our measurements?

Metric	Value
Measurements compared	360
Bias (simulated - measured)	-0.58 dB
Mean absolute error	3.79 dB
Correlation r	0.71
Within +/- 5 dB	73 %

Each field measurement is compared with the grid cell it falls in (median 16 m away), at the hour it was taken.

Caveat: this is an in-sample check - the model that produces the grid was trained on these same points. It measures the fidelity of the chain model -> 40 m grid -> GAMA, not generalisation. The generalisation figures are those of section 5: R2 0.14 / MAE 5.26 dB under buffered loo 300 m.



7. Limitations, stated plainly

- **THE MODEL DOES NOT BEAT ITS OWN PHYSICAL BASELINE.** An OLS regression on log(distance to road) - one variable, two parameters - scores R^2 0.200 against 0.137 for the six-variable LightGBM under the reference protocol, and holds 0.189 against 0.029 under leave-one-site-out. Morphology aggregated over 300 m adds no measurable value over that single term. The LightGBM leads only under the most permissive split (600 m blocks). This is reported as a result, not hidden.
- **ABSOLUTE CALIBRATION.** The three phones are calibrated against each other, never against a reference instrument. A bias common to all three (+3.0 to +7.3 dB, bounded by anchoring on instrumented Vietnamese campaigns) is invisible in our data. The field campaign is closed and this cannot be repaired retrospectively. Contrasts are supported; absolute levels are indicative.
- **MEASUREMENT QUANTITY.** A 20-30 s sample is not the L_{Aeq} of any regulatory reference period. A single horn burst moves it by several dB - horn events reach +17 dB in Vietnamese traffic. This variance is a property of the quantity and caps the R^2 any spatial model can reach.
- **GENERALISATION.** Leave-one-site-out is negative on two of three sites (section 5.3): the model does not extrapolate to an unsampled typology. Maps are clipped to the sampled envelope.
- **SPATIAL RESOLUTION.** Features are aggregated over a 300 m radius, so adjacent 40 m cells share more than 98% of their disc. The map cannot resolve the facade/courtyard contrast (10-15 dB in dense fabric): predicted levels are visibly flatter than measured ones.
- **TEMPORAL COVERAGE.** Night (21:00-06:00) holds ~3% of measurements and nothing between 00:00 and 05:00, although that is the period with the strictest threshold. One season only.
- **TRAFFIC VIDEOS.** Per-frame vehicle DENSITY is not FLOW, and the detector was never validated against manual counts. Modal splits are lower bounds on motorcycle share (section 7).

8. Next steps

- **HYBRID MODEL.** Add a physical propagation core (CNOSSOS-EU via NoiseModelling, OSM inputs) and train the statistical model on the RESIDUAL. This restores the fine spatial contrast the 300 m aggregation destroys, and makes the map extrapolable, since physics does not depend on our sample.
- **TRAFFIC.** Re-derive flow and speed from the videos by object tracking and line crossing, and validate the detector against manual counts before any acoustic use.
- **DATA PUBLICATION.** Deposit measurements, forms and cleaning code with a DOI; add an ethics statement covering the public-space video recordings.
- **MANUSCRIPT.** Methods, the three negative results as contributions, discussion.

9. Summary

363 smartphone measurements across 3 contrasting districts document a consistent exposure pattern in space and time. A LightGBM trained directly on them reaches R^2 0.14 [-0.10, 0.33] / MAE 5.26 dB under buffered loo 300 m - but an OLS regression on log(distance to road) alone reaches 0.20. Three negative results are reported as contributions: a one-variable physical baseline beats the six-variable model and 300 m morphology adds nothing over it; cross-city transfer from Uganda fails even with identical instruments and invariant features; and per-frame vehicle density carries no recoverable acoustic signal. We claim contrasts, not certified absolute levels.